UNIT 4 · TOPIC 2(D)

Ignition, Starting and Troubleshooting

Syllabus topic: Various systems of IC engines: operation, adjustment and troubleshooting

Ignition (SI engines)

Battery-coil ignition — the current path:

1. Battery
2. Ignition coil primary
3. Contact breaker or electronic trigger
4. HT secondary
5. Distributor
6. Spark plugs (gap ~**0.6–0.8 mm**)

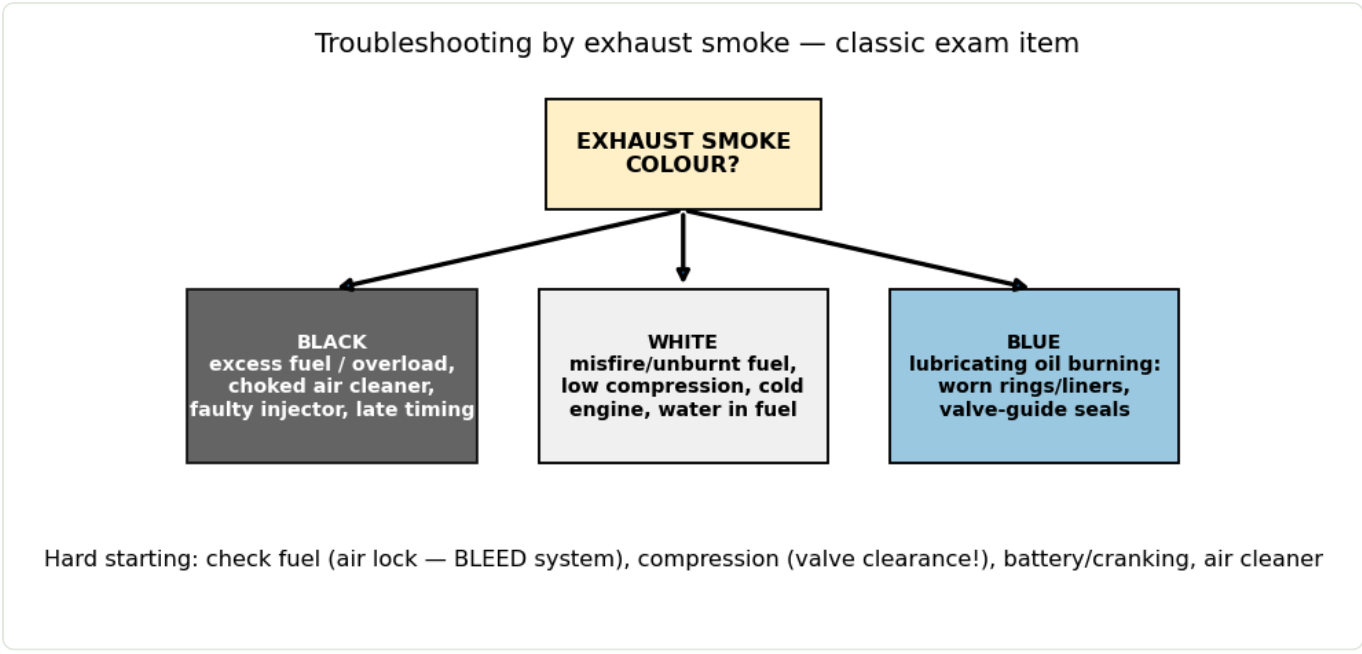
MAGNETO: self-generating — for small engines without a battery.

Diesel starting

- A diesel needs no spark, but uses an electric STARTING system: battery (Unit 3, Topic 12c), starter motor with Bendix/solenoid pinion, charging by alternator + regulator.
- Cold-start aids: glow plugs, intake heaters.

Troubleshooting summary

Symptom	Check
Hard starting (diesel)	Air in fuel lines (bleed); choked filters; low cranking speed (battery); low compression (valve clearance, worn rings); faulty injectors/timing
LOW POWER	Choked air cleaner; wrong timing; worn FIP/injectors; low compression; overheating
KNOCK	Diesel knock (long delay — low cetane, cold engine); mechanical knock (bearings, piston slap)
Excessive vibration	Misfiring cylinder (Morse-style cut-out check); loose mounts



Troubleshooting by exhaust smoke colour

Smoke	Meaning
BLUE	Burning lubricating oil
BLACK	Fuel-rich / air-starved
WHITE	Unburnt fuel / misfire / water

Practice MCQs

Q64. Blue exhaust smoke indicates:

(a) Excess fuel
(b) Burning lubricating oil (worn rings/guides)
(c) Water in fuel
(d) Retarded spark

Q65. White smoke from a diesel commonly indicates:

(a) Overload
(b) Misfire/unburnt fuel or water — low compression, cold start
(c) Oil burning
(d) Rich mixture

Q66. Black smoke under load indicates:

(a) Excess fuel/insufficient air — overload, choked air cleaner, bad injector
(b) Oil
(c) Coolant
(d) Normal

Q67. A magneto ignition system:

(a) Needs a battery
(b) Generates its own ignition current
(c) Works only on diesels
(d) Uses glow plugs

Q68. Spark plug gap is typically:

(a) 0.06–0.08 mm
(b) 0.6–0.8 mm
(c) 6–8 mm
(d) 2–3 mm

Q69. Cold-start aid in diesel engines:

(a) Choke
(b) Glow plugs / intake heater
(c) Carburettor
(d) Rich jet

Q70. Diesel knock is reduced by:

(a) Lower cetane fuel

(b) Higher cetane fuel, warm engine, higher compression temperature

(c) Bigger fan

(d) Lean mixture

Answers

Q64 — (b) Burning lubricating oil (worn rings/guides) — *Smoke triage: BLUE = oil, BLACK = fuel-rich/air-starved, WHITE = unburnt/misfire/water.*

Q65 — (b) Misfire/unburnt fuel or water — low compression, cold start

Q66 — (a) Excess fuel/insufficient air — overload, choked air cleaner, bad injector

Q67 — (b) Generates its own ignition current

Q68 — (b) 0.6–0.8 mm

Q69 — (b) Glow plugs / intake heater

Q70 — (b) Higher cetane fuel, warm engine, higher compression temperature — *Higher cetane shortens the ignition delay, so less fuel accumulates before first ignition — the opposite of SI knock chemistry (Topic 1b).*

Sources: Liljedahl et al. — Tractors and their Power Units; Jagdishwar Sahay; manufacturer service practice; TREM notifications; NPTEL farm power lectures.